

Multiple Choice (10 marks)

1. Which of the following is true for Br_2 at standard temperature and pressure?
 - (a) It is a colorless gas.
 - (b) It is a red-brown volatile liquid.
 - (c) It is a colorless volatile liquid.
 - (d) It is a yellow metallic solid.
2. A buffer is made from equal concentrations of a weak acid and its conjugate base. Doubling the volume of the buffer solution by adding water has what effect on its pH?
 - (a) It has little effect.
 - (b) It significantly increases the pH.
 - (c) It significantly decreases the pH.
 - (d) It changes the pH asymptotically to the pK_a of the acid.
3. ^{23}Na has an NMR frequency of 198.52 MHz in a 750 MHz NMR spectrometer. What is its magnetogyric ratio?
 - (a) $101.1 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$
 - (b) $26.75 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$
 - (c) $7.081 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$
 - (d) $0.9377 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$
4. A substance that is NOT generally considered to be a toxic pollutant in water is
 - (a) carbonic acid
 - (b) a halogenated hydrocarbon
 - (c) lead
 - (d) mercury
5. Which of the following is a primary standard for use in standardizing bases?
 - (a) Ammonium hydroxide
 - (b) Sulfuric acid
 - (c) Acetic acid
 - (d) Potassium hydrogen phthalate

True / False (10 marks)

Answer True or False

6. True or False: H_3O^+ and H_2O constitute a conjugate acid-base pair, where H_3O^+ is the conjugate acid of H_2O .
7. True or False: The molecular geometry of thionyl chloride, SOCl_2 , is best described as trigonal pyramidal due to its lone pair and three bonding pairs.
8. True or False: The infrared-active normal modes of a carbon dioxide molecule include symmetric stretching and bending.
9. True or False: The solid-state structures of the principal allotropes of elemental boron are made up of B_8 cubes.
10. True or False: The strongest base in liquid ammonia is the amide ion (NH_2^-).

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between pH and chemical shifts in the ^{31}P NMR spectrum of sodium phosphate. Specifically, analyze the expected chemical shift when the pH equals the pK_a of H_2PO_4^- .
12. Discuss the factors that determine the pH of 0.1 M aqueous solutions of various salts, and analyze why NaCl yields the lowest pH compared to other salts.

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